

IIE6120 Linear Programming
2025-2 Homework 05
Team B (이정민, 장서현)

[5.35] Let $A = \begin{bmatrix} 1 & 1 \\ 0 & 2 \\ -1 & 4 \end{bmatrix}$ and $c = (-1, 5)$. Which of the following two systems has

a solution?

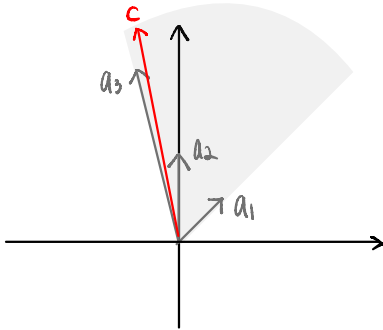
System 1: $Ax \leq 0, \quad cx > 0;$

System 2: $wA = c \quad w \geq 0.$

Illustrate geometrically.

#1 [5.35]

System 2)

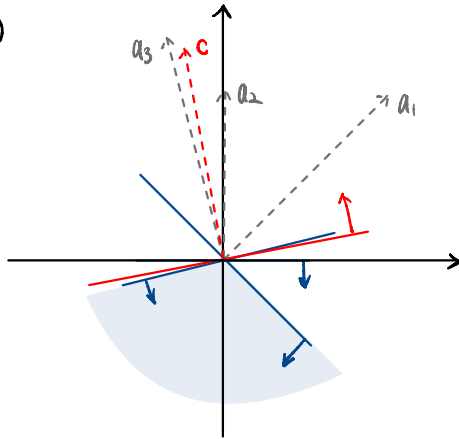


c lies within the cone generated by the rows of A (a_1, a_2, a_3).
 $\rightarrow c$ can be expressed as a non-negative linear combination of a_1, a_2, a_3 .

$\therefore$ System 2 has a solution.

By Farkas' Lemma, System 1 has no solution.

System 1)



The cone defined by $Ax \leq 0$ and the open half-space $cx > 0$ do not intersect.

$\therefore$ System 1 has no solution

[5.40] Consider the following problem:

$$\begin{array}{ll} \text{Maximize} & x_1 + 2x_2 \\ \text{subject to} & -2x_1 + x_2 \leq 2 \\ & x_2 \leq 4 \\ & x_1 + x_2 \leq 5 \\ & x_1, x_2 \geq 0. \end{array}$$

Starting at the origin, solve the problem by the simplex method, and exhibit that the optimal solution satisfies the KKT conditions. At each iteration, point out the source of violation of the optimality conditions. Interpret the KKT conditions geometrically at each extreme point encountered.

#2 [5.40]

$$\begin{array}{ll} \text{Min} & -x_1 - 2x_2 \quad C = [-1, -2] \\ \text{s.t.} & -2x_1 + x_2 + x_3 = 2 \\ & x_1 + x_2 + x_4 = 5 \\ & x_1, x_3, x_4 \geq 0 \\ & 0 \leq x_2 \leq 4 \end{array}$$

—————> initial BFS

$$\rightarrow x_1=0, x_2=0, x_3=2, x_4=5$$

(l) (l)

	z	x_1^l	x_2^l	x_3	x_4	RHS
z	1	1	2	0	0	0
x_3	0	-2	①	1	0	2
x_4	0	1	1	0	1	5

x_2 enter.

$$\Delta = \begin{cases} 4-2=2 & =2 \\ (2-0)/1=2 \\ (5-0)/1=5 \end{cases}$$

$$\begin{bmatrix} x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 2 \\ 5 \end{bmatrix} - \begin{bmatrix} 1 \\ 1 \end{bmatrix} \cdot 2 = \begin{bmatrix} 0 \\ 3 \end{bmatrix} \rightarrow x_3 \text{ leave (l)}$$

$$z = 0 - 2 \cdot 2 = -4, \quad x_2 = 0 + 2 = 2$$

KKT violation

: dual feasibility 위반

$$(z_1 - C_1 \geq 0, z_2 - C_2 \geq 0)$$

	z	x_1^l	x_2	x_3^l	x_4	RHS
z	1	5	0	-2	0	-4
x_2	0	-2	1	1	0	2
x_4	0	③	0	-1	1	3

x_1 enter

$$\Delta = \begin{cases} \infty & =1 \\ (4-2)/2=1 \\ (3-0)/3=1 \end{cases} > \text{tie} \rightarrow \text{degeneracy}$$

$$\begin{bmatrix} x_2 \\ x_4 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \end{bmatrix} - \begin{bmatrix} -2 \\ 3 \end{bmatrix} = \begin{bmatrix} 4 \\ 0 \end{bmatrix} \rightarrow x_4 \text{ leave (l)}$$

$$z = -4 - 5 \cdot 1 = -9, \quad x_1 = 0 + 1 = 1$$

KKT violation

: dual feasibility 위반

$$(z_1 - C_1 \geq 0)$$

	z	x_1	x_2	x_3^l	x_4^l	RHS
z	1	0	0	-1/3	-5/3	-9 (min 1/2)
x_2	0	0	1	1/3	2/3	4
x_1	0	1	0	-1/3	1/3	1

optimal $z^* = 9$ (max 1/2)

$$x_1 = 1, x_2 = 4$$

KKT condition satisfied

1) Primal feasibility satisfied

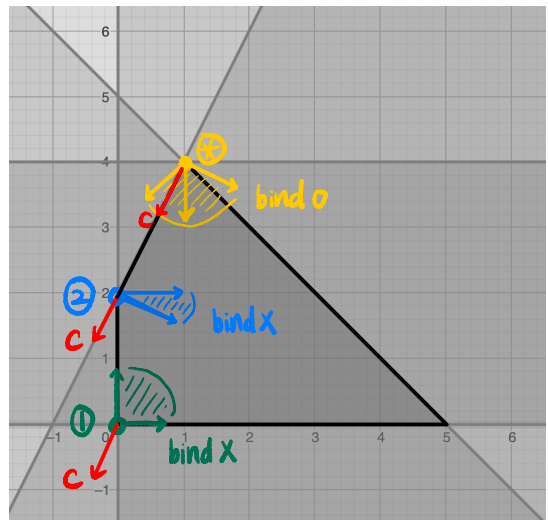
2) Dual feasibility satisfied

3) Comp. slackness satisfied.

$$Ax = b, X \geq 0$$

$$wA + v = C, w \text{ urs. } v \geq 0$$

$$w(Ax - b) = 0, vX = 0$$



[5.41] Write the KKT optimality conditions for each of the following problems:

c. Minimize $\mathbf{c}\mathbf{x}$
 subject to $\mathbf{A}\mathbf{x} \leq \mathbf{b}$
 $\mathbf{x} \geq \mathbf{0}$.

d. Minimize $\mathbf{c}\mathbf{x}$
 subject to $\mathbf{A}_1\mathbf{x} = \mathbf{b}_1$
 $\mathbf{A}_2\mathbf{x} \geq \mathbf{b}_2$
 $\mathbf{x} \geq \mathbf{0}$.

e. Minimize $\mathbf{c}\mathbf{x}$
 subject to $\mathbf{A}\mathbf{x} = \mathbf{b}$
 $\ell \leq \mathbf{x} \leq \mathbf{u}$.

#3 [5.41]

(c)

$w: \mathbf{A}\mathbf{x} \leq \mathbf{b}$ 의 dual, $v: \mathbf{x} \geq \mathbf{0}$ 의 dual

- Primal Feasibility : $\mathbf{A}\mathbf{x} \leq \mathbf{b}, \mathbf{x} \geq \mathbf{0}$
- Dual Feasibility : $\mathbf{C} = -\mathbf{w}\mathbf{A} + \mathbf{v}, \mathbf{w} \geq \mathbf{0}, \mathbf{v} \geq \mathbf{0}$
- Complementary Slackness : $\mathbf{w}(-\mathbf{A}\mathbf{x} + \mathbf{b}) = \mathbf{0}, \mathbf{v}\mathbf{x} = \mathbf{0}$.

(d)

$w_1: \mathbf{A}_1\mathbf{x} = \mathbf{b}_1$ 의 dual, $w_2: \mathbf{A}_2\mathbf{x} \geq \mathbf{b}_2$ 의 dual, $v: \mathbf{x} \geq \mathbf{0}$ 의 dual

- Primal Feasibility : $\mathbf{A}_1\mathbf{x} = \mathbf{b}_1, \mathbf{A}_2\mathbf{x} \geq \mathbf{b}_2, \mathbf{x} \geq \mathbf{0}$
- Dual Feasibility : $w_1\mathbf{A}_1 + w_2\mathbf{A}_2 + \mathbf{v} = \mathbf{C}, w_1$ unrestricted, $w_2 \geq \mathbf{0}, \mathbf{v} \geq \mathbf{0}$.
- Complementary Slackness : $w_1(\mathbf{A}_1\mathbf{x} - \mathbf{b}_1) = \mathbf{0}, w_2(\mathbf{A}_2\mathbf{x} - \mathbf{b}_2) = \mathbf{0}, \mathbf{v}\mathbf{x} = \mathbf{0}$.

(e)

$w: \mathbf{A}\mathbf{x} = \mathbf{b}$ 의 dual, $v_\ell: \mathbf{x} \geq \ell$ 의 dual, $v_u: \mathbf{x} \leq u$ 의 dual

- Primal Feasibility : $\mathbf{A}\mathbf{x} = \mathbf{b}, \ell \leq \mathbf{x} \leq \mathbf{u}$
- Dual Feasibility : $\mathbf{W}\mathbf{A} + \mathbf{v}_\ell - \mathbf{v}_u = \mathbf{C}, w$ unrestricted, $\mathbf{v}_\ell \geq \mathbf{0}, \mathbf{v}_u \geq \mathbf{0}$
- Complementary Slackness : $\mathbf{W}(\mathbf{A}\mathbf{x} - \mathbf{b}) = \mathbf{0}, \mathbf{v}_\ell(\mathbf{x} - \ell) = \mathbf{0}, \mathbf{v}_u(\mathbf{u} - \mathbf{x}) = \mathbf{0}$

[5.43] Consider the problem: $\textcircled{P_0}$ Minimize cx subject to $Ax \geq b$, $x \geq 0$. Let x^* be an optimal solution. Suppose that A is decomposed into $\begin{bmatrix} A_1 \\ A_2 \end{bmatrix}$ and b is decomposed into $\begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$ such that $A_1 x^* = b_1$ and $A_2 x^* > b_2$. Show that x^* is also an optimal solution to the problem: $\textcircled{P_1}$ Minimize cx subject to $A_1 x \geq b_1$, $x \geq 0$, and to the problem: $\textcircled{P_2}$ Minimize cx subject to $A_1 x = b_1$, $x \geq 0$.

#4 [5.43]

(P₀) Min $cx \rightarrow x^*$: optimal solution

s.t. $Ax \geq b \rightarrow w^* = (w_1^*, w_2^*) \geq 0 \rightarrow$ Due to complementary slackness,

$x \geq 0 \rightarrow v^* \geq 0$

$$\begin{cases} A_1 x^* = b_1 & : \text{binding} \\ A_2 x^* > b_2 & : \text{non-binding} \rightarrow w_2^* = 0 \\ x^* \geq 0 & \rightarrow v^* x^* = 0 \end{cases}$$

Due to dual feasibility, $w^* A + v^* = c$

$$[w_1^*, 0] \begin{bmatrix} A_1 \\ A_2 \end{bmatrix} + v^* = c$$

$$w_1^* A_1 + v^* = c, \quad w_1^* \geq 0, v^* \geq 0.$$

(P₁) Min $cx \rightarrow x^*$

s.t. $A_1 x \geq b_1 \rightarrow w_1^*$

$x \geq 0 \rightarrow v^*$

1) Primal Feasibility : Satisfied

Since $A_1 x^* = b_1$, $A_1 x^* \geq b_1$ holds. $x^* \geq 0$.

2) Dual Feasibility : Satisfied

$w_1^* A_1 + v^* = c$, $w_1^* \geq 0$, $v^* \geq 0$.

3) Complementary Slackness : Satisfied

$w_1^* (A_1 x^* - b_1) = 0$, $v^* x^* = 0$

Since x^* satisfies all KKT conditions,
 x^* is an optimal solution for P₁

(P₂) Min $cx \rightarrow x^*$

s.t. $A_1 x = b_1 \rightarrow w_1^*$

$x \geq 0 \rightarrow v^*$

1) Primal Feasibility : Satisfied

$A_1 x^* = b_1$, $x^* \geq 0$.

2) Dual Feasibility : Satisfied

$w_1^* A_1 + v^* = c$, $v^* \geq 0$. w^* unrestricted ($w^* \geq 0$ included)

3) Complementary Slackness : Satisfied

$w_1^* (A_1 x^* - b_1) = 0$, $v^* x^* = 0$

Since x^* satisfies all KKT conditions,
 x^* is an optimal solution for P₂

[6.6] Consider the following problem:

$$\begin{array}{llllll} \text{Minimize} & 2x_1 & + & 15x_2 & + & 5x_3 & + & 6x_4 \\ \text{subject to} & x_1 & + & 6x_2 & + & 3x_3 & + & x_4 & \geq & 2 \\ & -2x_1 & + & 5x_2 & - & 4x_3 & + & 3x_4 & \leq & -3 \\ & x_1, & & x_2, & & x_3, & & x_4 & \geq & 0. \end{array}$$

- Give the dual linear problem.
- Solve the dual geometrically.
- Utilize information about the dual linear program and the theorems of duality to solve the primal problem.

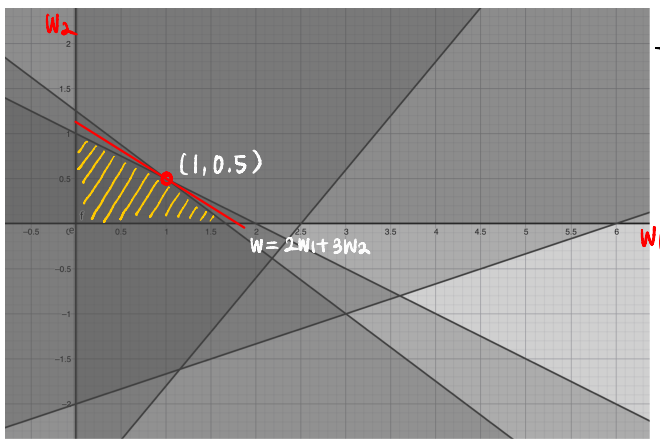
#5 [6.6]

a.

$$\begin{array}{ll} \text{(Primal)} & \text{Min } 2x_1 + 15x_2 + 5x_3 + 6x_4 \\ & \text{s.t. } x_1 + 6x_2 + 3x_3 + x_4 \geq 2 \\ & \quad 2x_1 - 5x_2 + 4x_3 - 3x_4 \geq 3 \\ & \quad x_1, x_2, x_3, x_4 \geq 0 \end{array}$$

$$\begin{array}{ll} \longleftrightarrow \text{(Dual)} & \text{Max } 2w_1 + 3w_2 \\ & \text{s.t. } w_1 + 2w_2 \leq 2 \quad \text{... ①} \\ & \quad 6w_1 - 5w_2 \leq 15 \quad \text{... ②} \\ & \quad 3w_1 + 4w_2 \leq 5 \quad \text{... ③} \\ & \quad w_1 - 3w_2 \leq 6 \quad \text{... ④} \\ & \quad w_1, w_2 \geq 0 \end{array}$$

b.



$$\begin{array}{l} \rightarrow w_1 = 1, w_2 = 0.5 \\ w^* = 3.5 \end{array}$$

c.

$$1) z^* = w^* = 3.5$$

Since the dual has an optimal solution, the primal also has an optimal solution.
 $\rightarrow$ Their objective values are equal.

2) Using Complementary Slackness Theorem,

- Since $w_1 > 0$ and $w_2 > 0$, $\begin{cases} x_1 + 6x_2 + 3x_3 + x_4 = 2 \\ 2x_1 - 5x_2 + 4x_3 - 3x_4 = 3 \end{cases}$ must hold.

- Using $w_1 = 1, w_2 = 0.5$, $\begin{cases} \text{constraints ① ③ are binding} \rightarrow x_1, x_3 \text{ can be positive} \\ \text{constraints ② ④ has slack} \rightarrow x_2 = 0, x_4 = 0. \end{cases}$

- $\begin{cases} x_1 + 3x_3 = 2 \\ 2x_1 + 4x_3 = 3 \end{cases} \rightarrow x_1 = \frac{1}{2}, x_3 = \frac{1}{2}$

$$\begin{array}{l} \therefore z^* = 3.5, \\ x_1 = \frac{1}{2}, x_2 = 0, x_3 = \frac{1}{2}, x_4 = 0. \end{array}$$

[6.13] Consider the problem: Minimize $\mathbf{c}\mathbf{x}$ subject to $\mathbf{A}\mathbf{x} = \mathbf{b}$, $\mathbf{x} \geq \mathbf{0}$ where $m = n$, $\mathbf{c} = \mathbf{b}^t$ and $\mathbf{A} = \mathbf{A}^t$. Show that if there exists an $\mathbf{x}_0$ such that $\mathbf{A}\mathbf{x}_0 = \mathbf{b}$, $\mathbf{x}_0 \geq \mathbf{0}$, then $\mathbf{x}_0$ is an optimal solution. (Hint: Use duality.)

#6 [6.13]

$$\begin{array}{ll} \text{(Primal)} \quad \text{Min } z = \mathbf{c}\mathbf{x} & \longleftrightarrow \quad \text{(Dual)} \quad \text{Max } v = \mathbf{w}\mathbf{b} \\ \text{s.t. } \mathbf{A}\mathbf{x} = \mathbf{b} & \text{s.t. } \mathbf{w}\mathbf{A} \leq \mathbf{c} \\ \mathbf{x} \geq \mathbf{0} & \mathbf{w} \text{ unrestricted.} \end{array}$$

Given conditions $\left. \begin{array}{l} \textcircled{1} m=n \\ \textcircled{2} \mathbf{A} = \mathbf{A}^t \\ \textcircled{3} \mathbf{c} = \mathbf{b}^t \\ \textcircled{4} \mathbf{A}\mathbf{x}_0 = \mathbf{b}, \mathbf{x}_0 \geq \mathbf{0} \end{array} \right\} \mathbf{A} : \text{symmetric, square}$

Let $\mathbf{w}_0$ be a dual solution based on the primal solution $\mathbf{x}_0$. $\rightarrow \mathbf{w}_0 = \mathbf{x}_0^t$

1) Verify $\mathbf{w}_0$'s dual feasibility

$\mathbf{w}\mathbf{A} \leq \mathbf{c} \rightarrow \mathbf{x}_0^t \mathbf{A} \leq \mathbf{c}$ needs to hold

$$\begin{aligned} \text{transpose of LHS: } (\mathbf{x}_0^t \mathbf{A})^t &= \mathbf{A}^t \mathbf{x}_0 = \mathbf{A} \mathbf{x}_0 = \mathbf{b} \\ \mathbf{x}_0^t \mathbf{A} &= \mathbf{b}^t = \mathbf{c} \end{aligned}$$

Thus, $\mathbf{w}_0 = \mathbf{x}_0^t$ is a feasible solution to the dual

2) Verify strong duality

$$z_0 = \mathbf{c}\mathbf{x}_0$$

$$= \mathbf{b}^t \mathbf{x}_0$$

$$= \mathbf{x}_0^t \mathbf{b}$$

$$= \mathbf{w}_0 \mathbf{b}$$

$$= v_0$$

$\rightarrow$ primal & dual problem's objective value are equal $z_0 = v_0$

According to Strong Duality, both solutions $\mathbf{x}_0$ & $\mathbf{w}_0$ are optimal for their respective problems.

$\therefore \mathbf{x}_0$ is an optimal solution.

[6.28] The following is an optimal simplex tableau for a maximization problem having all $\leq$ constraints:

		SLACKS						
	z	x_1	x_2	x_3	x_4	x_5	x_6	RHS
z	1	0	0	0	3	0	5	θ
x_1	0	1	1	0	2	0	1	2
x_3	0	0	0	1	1	0	4	2
x_5	0	0	-2	0	-1	1	3	1

- Give the optimal solution.
- Give the optimal dual solution.
- Find $\partial z / \partial b_1$. Interpret this number.

- Find $\partial x_1 / \partial x_6$. Interpret this number.
- If you could buy an additional unit of the first resource for a cost of $5/2$, would you do this? Why?
- Another firm wishes to purchase one unit of the third resource from you. How much is such a unit worth to you? Why?
- Are there any alternative optimal solutions? If not, why not? If so, give one.
- Find the objective value θ .

#7 [6.28]

a. $x_1 = 2, x_2 = 0, x_3 = 2$
(slack: $x_4 = 0, x_5 = 1, x_6 = 0$)

b. $w_1 = 3, w_2 = 0, w_3 = 5$

c. $\partial z / \partial b_1 = 3$

An increase in b_1 (RHS of constraint 1) leads to the increase of z^* (objective value) by 3.

d. $\partial x_1 / \partial x_6 = -1$

An increase in x_6 forces x_1 to decrease by 1 to maintain feasibility

e. Yes.

Shadow price of resource 1 = 3
cost of buying = 2.5
profit = $3 - 2.5 = 0.5 > 0$.

f. 5

Shadow price of resource 3 = 5
→ One unit of resource 3 increases my profit by 5

g. Alternative optimal solutions exist. → $z_2 - c_2 = 0$ when x_2 is a NBV.

$$\begin{bmatrix} x_1 \\ x_3 \\ x_5 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \\ 1 \end{bmatrix} - 2 \cdot \begin{bmatrix} 1 \\ 0 \\ -2 \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \\ 5 \end{bmatrix}$$

$$x_2 = 0 + 2 = 2 \rightarrow (x_1, x_2, x_3, x_4, x_5, x_6) = (0, 2, 2, 0, 5, 0)$$

More generally,

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \\ x_6 \end{pmatrix} = \begin{pmatrix} 2 \\ 0 \\ 2 \\ 0 \\ 1 \\ 0 \end{pmatrix} \lambda_1 + \begin{pmatrix} 0 \\ 2 \\ 2 \\ 0 \\ 5 \\ 0 \end{pmatrix} \lambda_2$$

where $\lambda_1 + \lambda_2 = 1$,
 $\lambda_1, \lambda_2 \geq 0$.

h. $\theta = 4$

$$\begin{aligned} \bullet z_5 - c_5 &= 0, \quad c_B B^{-1} a_5 - c_5 = c_B y_5 - c_5 \\ &= [c_1, c_3, c_5] \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = c_5 = 0 \end{aligned}$$

$$\bullet z_4 - c_4 = 3, \quad [c_1, c_3, 0] \begin{bmatrix} 2 \\ 1 \\ -1 \end{bmatrix} = 2c_1 + c_3 = 3$$

$$\bullet z_6 - c_6 = 5, \quad [c_1, c_3, 0] \begin{bmatrix} 1 \\ 4 \\ 3 \end{bmatrix} = c_1 + 4c_3 = 5$$

$$\rightarrow c_1 = 1, c_3 = 1$$

$$\hookrightarrow c_B = [c_1, c_3, c_5] = [1, 1, 0]$$

$$z^* = \theta = c_B B^{-1} b = [1, 1, 0] \begin{bmatrix} 2 \\ 2 \\ 1 \end{bmatrix} = 4$$

[6.30] Solve the following linear program by the dual simplex method:

$$\begin{array}{llllll} \text{Minimize} & 2x_1 & + & 3x_2 & + & 5x_3 & + & 6x_4 \\ \text{subject to} & 2x_1 & + & 2x_2 & + & 3x_3 & + & x_4 & \geq & 3 \\ & -x_1 & + & x_2 & - & x_3 & + & 3x_4 & \leq & -3 \\ & x_1, & & x_2, & & x_3, & & x_4 & \geq & 0. \end{array}$$

#8 [6.30]

$$\text{Min } 2x_1 + 3x_2 + 5x_3 + 6x_4$$

$$\text{s.t. } 2x_1 + 2x_2 + 3x_3 + x_4 \geq 3$$

$$-x_1 + x_2 - x_3 + 3x_4 \leq -3$$

$$x_1, x_2, x_3, x_4 \geq 0$$

$$\longrightarrow \text{Min } 2d_1 + 3d_2 + 5d_3 + 6d_4$$

$$\text{s.t. } 2d_1 + 2d_2 + 3d_3 + d_4 - d_5 = 3$$

$$-d_1 + d_2 - d_3 + 3d_4 + d_6 = -3$$

$$d_1, d_2, d_3, d_4, d_5, d_6 \geq 0.$$

primal optimality satisfied

	z	x_1	x_2	x_3	x_4	x_5	x_6	RHS
z	1	-2	-3	-5	-6	0	0	0
x_5	0	-2	-2	-3	-1	1	0	-3
x_6	0	-1	1	-1	3	0	1	-3

→ primal feasibility unsatisfied.
→ x_6 leave, x_1 enter

	z	x_1	x_2	x_3	x_4	x_5	x_6	RHS
z	1	0	-5	-3	-12	0	-2	6
x_5	0	0	-4	-1	-7	1	-2	3
x_1	0	1	-1	1	-3	0	-1	3

→ primal optimality (= dual feasibility) satisfied.
→ primal feasibility satisfied.

$$\therefore z^* = 6$$

$$d_1 = 3, d_2 = d_3 = d_4 = 0$$

$$(\text{slack: } d_5 = 3, d_6 = 0)$$

[6.54] Consider the following linear programming problem and its optimal final tableau shown below:

$$\begin{aligned} \text{Maximize} \quad & 2x_1 + x_2 - x_3 \\ \text{subject to} \quad & x_1 + 2x_2 + x_3 \leq 8 \\ & -x_1 + x_2 - 2x_3 \leq 4 \\ & x_1, x_2, x_3 \geq 0. \end{aligned}$$

Final Tableau:

	z	x_1	x_2	x_3	x_4	x_5	RHS
z	1	0	3	3	2	0	16
x_1	0	1	2	1	1	0	8
x_5	0	0	3	-1	1	1	12

- Write the dual problem and give the optimal dual variable values from the foregoing tableau.
- Using sensitivity analysis, find a new optimal solution if the coefficient of x_2 in the objective function is changed from 1 to 5.
- Suppose that the coefficient of x_2 in the first constraint is changed from +2 to 1/6. Using sensitivity analysis, find a new optimal solution.
- Suppose that the following constraint is added to the problem: $x_2 + 2x_3 = 3$. Using sensitivity analysis, find the new optimal solution.
- If you were to choose between increasing the right-hand-side of the first and second constraints, which one would you choose? Why? What is the effect of this increase on the optimal value of the objective function?
- Suppose that a new activity x_6 is proposed with unit return 6 and consumption vector $a_6 = (2, 1)^T$. Find a new optimal solution.

#9 [6.54]

$$\begin{aligned} \text{(a) Min } & 8w_1 + 4w_2 \\ \text{s.t. } & w_1 - w_2 \geq 2 \\ & 2w_1 + w_2 \geq 1 \\ & w_1 - 2w_2 \geq -1 \\ & w_1, w_2 \geq 0 \end{aligned}$$

$$w_1^* = 2, w_2^* = 0$$

$$\text{(b) } (x_1, x_2, x_3) = (0, 4, 0), z^* = 20$$

$$x_2 \text{ is non-basic. Thus, } z_2 - c_2 \text{ is replaced by } (z_2 - c_2) - (c_2 - c_2') = 3 - (5 - 1) = -1$$

Since $z_2 - c_2 \leq 0$, x_2 must enter.

Minimum ratio test results in a tie ($=4$), so we arbitrarily choose x_1 as the leaving variable.

$$z = 16 - (-1) \cdot 4 = 20$$

$$\begin{bmatrix} x_1 \\ x_5 \end{bmatrix} = \begin{bmatrix} 8 \\ 12 \end{bmatrix} - \begin{bmatrix} 2 \\ 3 \end{bmatrix} \cdot 4 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, x_2 = 0 + 4 = 4$$

$$\text{(c) } (x_1, x_2, x_3) = (44/7, 72/7, 0), z^* = 160/7$$

$$y_2 = B^{-1}a_2' = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1/6 \\ 1 \end{bmatrix} = \begin{bmatrix} 1/6 \\ 7/6 \end{bmatrix}$$

$$\begin{aligned} z_2 - c_2 &= c_B B^{-1} a_2' - c_2 \\ &= w a_2' - c_2 = \begin{bmatrix} 2 & 0 \end{bmatrix} \begin{bmatrix} 1/6 \\ 7/6 \end{bmatrix} - 1 = -2/3 \end{aligned}$$

x_2 enters, x_5 leaves ($\theta = 72/7$)

$$z = 16 - (-2/3) \cdot 72/7 = 160/7$$

$$\begin{bmatrix} x_1 \\ x_5 \end{bmatrix} = \begin{bmatrix} 8 \\ 12 \end{bmatrix} - \begin{bmatrix} 1/6 \\ 7/6 \end{bmatrix} \cdot 72/7 = \begin{bmatrix} 44/7 \\ 0 \end{bmatrix}, x_2 = 0 + 72/7 = 72/7$$

$$\text{(d) } (x_1, x_2, x_3) = (6.5, 0, 1.5), z^* = 11.5$$

$$x_2 \text{ reduced cost} = 3, \text{ contribution to constraint} = 1 \rightarrow 3/1 = 3$$

$$x_3 \text{ reduced cost} = 3, \text{ contribution to constraint} = 2 \rightarrow 3/2 = 1.5$$

$$\rightarrow x_2 = 0, x_3 = 1.5$$

$$x_1 = 8 - 1.5 = 6.5, x_5 = 12 + 1.5 = 13.5$$

(e) Constraint 1.

• Shadow prize of constraint 1 = 2

• Shadow prize of constraint 2 = 0

→ The obj. function value will increase by 2 for every 1 unit increase in b_1

$$\text{(f) } (x_1, x_2, x_3, x_6) = (0, 0, 0, 4), z^* = 24$$

$$z_6 - c_6 = c_B B^{-1} a_6 - c_6 = \begin{bmatrix} 2 & 0 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix} - 6 = -2$$

$$y_6 = B^{-1}a_6 = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

x_6 enters, we arbitrarily choose x_1 as leaving variable ($\theta = 4$)

$$z = 16 - (-2) \cdot 4 = 24$$

$$\begin{bmatrix} x_1 \\ x_5 \end{bmatrix} = \begin{bmatrix} 8 \\ 12 \end{bmatrix} - \begin{bmatrix} 2 \\ 3 \end{bmatrix} \cdot 4 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, x_6 = 0 + 4 = 4$$

[6.67] Consider the following problem:

$$\begin{aligned} \text{Maximize } & 2x_1 + 4x_2 + 5x_3 \\ \text{subject to } & x_1 + x_2 + 2x_3 \leq 8 \\ & x_1 - x_2 + 3x_3 \leq 4 \\ & x_1, x_2, x_3 \geq 0. \end{aligned}$$

$$\begin{aligned} \rightarrow \text{Max } & 2x_1 + 4x_2 + 5x_3 \\ \text{s.t. } & x_1 + x_2 + 2x_3 + x_4 = 8 \\ & x_1 - x_2 + 3x_3 + x_5 = 4 \\ & x_1, x_2, x_3, x_4, x_5 \geq 0. \end{aligned}$$

- Find an optimal solution.
- Find a new optimal solution if the cost coefficient c_2 changes from 4 to -4.
- Determine optimal solutions as the cost coefficient c_2 varies over the entire real line $(-\infty, \infty)$.

#10 [6.67]

(a)

	Z	x_1	x_2	x_3	x_4	x_5	RHS
Z	1	-2	-4	-5	0	0	0
x_4	0	1	1	2	1	0	8
x_5	0	1	-1	3	0	1	4

x_3 enters, x_5 leaves

	Z	x_1	x_2	x_3	x_4	x_5	RHS
Z	1	-1/3	-11/3	0	0	5/3	20/3
x_4	0	1/3	5/3	0	1	-2/3	16/3
x_3	0	1/3	-1/3	1	0	1/3	4/3

x_2 enters, x_4 leaves

	Z	x_1	x_2	x_3	x_4	x_5	RHS
Z	1	4/5	0	0	11/5	-3/5	124/5
x_2	0	1/5	1	0	3/5	-2/5	16/5
x_3	0	2/5	0	1	1/5	1/5	12/5

x_5 enters, x_3 leaves

	Z	x_1	x_2	x_3	x_4	x_5	RHS
Z	1	2	0	3	4	0	32
x_2	0	1	1	2	1	0	8
x_5	0	2	0	5	1	1	12

$\rightarrow$ optimal $z^* = 32$, $x_1 = 0$, $x_2 = 8$, $x_3 = 0$.

(b)

$$C_B = [-4, 0] \rightarrow W = C_B B^{-1} = [-4, 0] \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} = [-4, 0]$$

$$z_1 - c_1 = [-4, 0] \begin{bmatrix} 1 \\ 1 \end{bmatrix} - 2 = -6, \quad z_3 - c_3 = [-4, 0] \begin{bmatrix} 2 \\ 3 \end{bmatrix} - 5 = -13, \quad z_4 - c_4 = [-4, 0] \begin{bmatrix} 1 \\ 0 \end{bmatrix} - 0 = -4$$

$\downarrow$
 x_3 enter.

	Z	x_1	x_2	x_3	x_4	x_5	RHS
Z	1	-6	0	-13	-4	0	-32
x_2	0	1	1	2	1	0	8
x_5	0	2	0	5	1	1	12

x_3 enter, x_5 leave

	Z	x_1	x_2	x_3	x_4	x_5	RHS
Z	1	-4/5	0	0	-11/5	13/5	-4/5
x_2	0	1/5	1	0	3/5	-2/5	16/5
x_3	0	2/5	0	1	1/5	1/5	12/5

x_4 enter, x_2 leave

	Z	x_1	x_2	x_3	x_4	x_5	RHS
Z	1	-1/3	7/3	0	0	5/3	20/3
x_4	0	1/3	5/3	0	1	-2/3	16/3
x_3	0	1/3	-1/3	1	0	1/3	4/3

x_1 enter, x_3 leave

	z	x_1	x_2	x_3	x_4	x_5	RHS
z	1	0	2	1	0	2	8
x_4	0	0	2	-1	1	-1	4
x_1	0	1	-1	3	0	1	4

→ optimal $z^* = 8$, $x_1 = 4$, $x_2 = 0$, $x_3 = 0$

(c)

$x_1 + x_2 + 2x_3 = 8$	...	①	(x_1, x_2, x_3)	$z(C_2)$	
$x_1 - x_2 + 3x_3 = 4$	...	②	①③⑤ → $(0, 8, 0)$	$8C_2$	] $C_2 = 2.5$] $C_2 = 0$] $C_2 = -2$
$x_1 = 0$	...	③	①②③ → $(0, 16/5, 12/5)$	$16/5 C_2 + 12$	
$x_2 = 0$	...	④	①②⑤ → $(6, 2, 0)$	$2C_2 + 12$	
$x_3 = 0$	...	⑤	②④⑤ → $(4, 0, 0)$	8	

$$x^* = (x_1^*, x_2^*, x_3^*) = \begin{cases} (0, 8, 0) & 2.5 \leq C_2 \\ (0, 16/5, 12/5) & 0 \leq C_2 < 2.5 \\ (6, 2, 0) & -2 \leq C_2 < 0 \\ (4, 0, 0) & C_2 < -2 \end{cases}$$